sustainability

MDPI

Article

Impact of Accessibility to Educational Facilities on Commuters' Transportation Choices: Married Females and School-Age Children

Wonchul Kim

Department of Spatial & Environmental Planning, Chungnam Institute, Gongju 32589, Republic of Korea; iwonchul@cni.re.kr

Abstract: This paper empirically analyzes the factors influencing the choice of transportation mode of commuters (especially married women), as well as the influence of accessibility to educational facilities on the choice of transportation mode of parents with school-age children. The results of this study suggest that improving public transportation and accessibility to educational facilities in the region will be important in relation to increasing public transportation usage. The longer it takes for students in a given area to commute to school using public transportation compared to driving, the more likely it is that parents will drive their children to or from school. An 8.4 min increase in elementary school students' commuting by public transit leads to a decrease in the probability for their parents to choose public transit in their commuting by 55 percentage points. In addition, when the students are younger, the probability of using public transportation is lower. Married females' education level and labor force participation have been increasing in Korea. Thus, their opportunity cost of driving their children to or from school rises and accessibility to educational facilities by public transit is more important than before.

Keywords: accessibility to school; parents' commuting mode; married female workers' child-rearing burden

check for updates

Academic Editors: Marilisa Botte and Giovanni Leonardi

Received: 12 March 2025
Revised: 8 April 2025
Accepted: 24 April 2025
Published: 26 April 2025

Citation: Kim, W. Impact of Accessibility to Educational Facilities on Commuters' Transportation Choices: Married Females and School-Age Children. *Sustainability* **2025**, *17*, 3920. https://doi.org/10.3390/su17093920

Copyright: © 2025 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

In Korea, as the economy has grown, women's educational levels have increased and their participation in economic activities has become more active. This has led to an increase in dual-income households; a change in the perception of the labor supply of married women who were previously considered as a secondary source of income; and a change in transportation modes for commuting. The active economic participation of women with higher levels of education is a desirable phenomenon in terms of human capital, not only for the individuals concerned, but also for society as a whole. However, the female labor force has been mentioned as one of the causes of low birth rates due to the increase in the age of first marriage and the delay in childbirth. Since the burden of childcare and housework on married women is still considerable, the accessibility of educational facilities for school-age children in local residential areas will affect married women's commuting time and their choice of commuting mode.

Couples form a household, spend a significant portion of their lives together, and engage in economic activities as a single household unit; therefore, they make optimal decisions regarding economic participation, income, and consumption. The commuting behavior of dual-income couples may not simply be a result of the income that can be earned through work, but may also be a result of comprehensive decision-making that

reflects the individual's human capital characteristics, family life cycle, class, and even discussions with the spouse. Children are also presented as a factor that significantly affects commuting, whereby the higher the number of children and the younger the children, the shorter the commuting time or distance [1–3]; the effect is presented to be more evident in women than in men [3,4].

Even though physical conditions have improved and the government has provided support to improve mobility by considering factors such as income and age, the commuting behavior of urban residents still appears to differ depending on factors such as gender, whether or not they use a private car, and where they live. Women, who have a low rate of economic activity participation and are responsible for housework and childcare, are assumed to have a commuting behavior centered around their residential areas; additionally, their preferred modes of transportation is assumed to be different to that of men, who mostly use their own cars rather than walking or using public transportation. However, beyond these assumptions, there is a lack of empirical discussion on the differences in gender mobility and the link between individual characteristics and mobility behavior characteristics. In particular, the worse the accessibility to educational facilities by public transportation compared to private vehicles for parents with school-age children, the less likely it is that parents in that city will commute by public transportation. In addition, because the burden of child-rearing differs between husbands and wives and the distances to schools differ between elementary-, middle-, and high-school-age children, there will be different outcomes between husbands and wives and between school-age children.

Therefore, this study empirically analyzes the factors influencing the choice of transportation mode of commuters (especially married women) and the accessibility to educational facilities on the choice of transportation mode of parents with school-age children using a binary logit model. The results of this study suggest that improving public transportation and accessibility to educational facilities in the region will be important for increasing public transportation usage. The data used in the empirical work include two percent samples of the Korean Population Census. The enrichment of the data allows us to control for the commuters' attributes in detail, mitigating possible concerns related to econometric issues.

The structure of this paper is as follows. Section 2 examines related prior research. Section 3 describes the data and the summary statistics of variables used in the empirical analysis and presents a binary logit model to empirically verify the effect of differences in accessibility to educational facilities within a city on the choice of commuting transportation mode; the results of the empirical analysis are discussed in Section 4. Section 5 discusses the academic and policy implications of the main results derived from the empirical analysis, while the Section 6 presents the conclusions, limitations of the study, and future tasks.

2. Literature Review

2.1. Factors Influencing Transportation Mode Choice Behavior

Transportation choice behavior is determined by a combination of various factors. Typically, three influencing factors, as highlighted by Ortuzar and Willumsen [5], are considered important—the individual characteristics, the trip characteristics, and the transport facility related to the built environment.

Regarding the individual characteristics, Truong and Somenahalli [6] argued that older adults who dislike driving during peak commuting hours tend to use public transportation. However, a study by Pettersson et al. [7] found that age played a negative role in avoiding public transport use among older people in Western European countries. Jin and Yu [8], who examined gender characteristics, presented research results showing that women over 40 years of age tend to use public transportation less than men. When considering both age

and income level, the elderly, students, and workers tend to use buses more [9]. Regarding working hours, Batabyal and Beladi [10] argued that extended working hours can increase the preference for using public transportation or private cars to quickly travel during rush hour when traffic congestion occurs. In households with only one car, the male's use of passenger cars increases to cope with long distances and work-related tasks [11,12].

Regarding the trip characteristics, which consist of commuting time, commuting distance, and trip chain complexity, the results showed that women's commuting distances are shorter than those of men, since women have more childcare and housework responsibilities than men [13,14]. Regarding the complexity of the trip chain, it was argued that women prefer various modes of transportation, such as combining walking with public transportation, within a single trip, whereas men prefer one mode of transportation, such as driving, for the entire trip [15].

Regarding the transport facility related to the built environment, Liu et al. [16] reported that in urban areas with a mix of commercial, residential, and educational facilities, students prefer active transport such as bicycles, whereas in suburban areas, they prefer cars. Wang et al. [17] suggested that the residential built environment is an important variable affecting commuting behavior for women; as concern men, population density, land use, and accessibility are more important variables affecting commuting behavior than the residential built environment. In addition, regarding the influence of the topographic factors of the residence on the choice of commuting transportation, Hayauchi et al. [18] emphasized that elevation differences in territory increase the preference for vehicle use such as buses, taxis, and passenger cars over walking. Sun et al. [19] found that road density, population density, and four-way intersections in residential areas were negatively related to the probability of driving alone.

2.2. Accessibility to Education Facility and Transportation Mode Choice

Considering accessibility to educational facilities, the main factors influencing general transportation mode selection behavior, i.e., the individual characteristics of the traveler, the trip characteristics, as well as the characteristics of the transport facility, play an important role in the commuter's transportation mode selection behavior. Specifically, the main variables that affect the choice of commuting transportation, considering the accessibility of educational facilities, were highlighted as access time to educational facilities, access distance to educational facilities, number of family members, income level, car ownership of family members, and perception of transportation, climate, and culture [20–22].

Zhang et al. [23], who emphasized that accessibility to educational facilities is an important variable affecting transportation mode selection behavior, analyzed the transportation mode selection behavior of households with school students aged between seven and eighteen in Beijing, China; they presented results that show that the possibility of choosing a passenger car increases when owning a passenger car or when the walking/cycling environment is not good. Furthermore, it was emphasized that this dependence on passenger cars increases when accessibility to educational facilities is poor, i.e., when the school is a long distance away. Similarly, Singh [24] found that when the public transportation service is poor or there is no school bus, the likelihood of using private cars or paratransit increases with regard to the choice of school travel mode. Ahern et al. [25], targeting Yorkshire, England, also emphasized that the distance, time constraints, and parent's safety concerns are important variables influencing the choice of commuting transportation considering the accessibility of educational facilities.

The most important factor to consider when analyzing the impact of accessibility to educational facilities on commuter's transportation choices is school-age children, who are the main users of educational facilities. Specifically, the presence or absence of school-age

children and the age of the school-age children are influential factors in parents' choice of transportation for commuting. Fitzhugh et al. [26] pointed out that children are a very important variable influencing parents' choice of transportation for commuting, especially for young couples, as they aim to ensure that the safety and convenience of their children are protected above all else. Specifically, Guthrie and Fan [27] emphasized that parents with young children are relatively less likely to use public transportation compared to parents without children. Compared to men, women face an increased burden of childcare after giving birth and are more likely to rely on private cars rather than public transportation to balance work and childcare, leading to a wider gap in the choice of commuting transportation [28–30]. Borghorst et al. [31] empirically showed that women with children are much more likely to leave their job when they have a long commute. Furthermore, it was emphasized that as the number of family members increases, the dependence on passenger cars in such households increases to meet the daily transportation needs of many family members [32].

3. Methodology

3.1. Data

For the empirical analysis of this paper, 2020 Population and Housing Census data from Statistics Korea, as well as the 2020 transportation accessibility index from the Korea Transport Institute (KOTI), are used. The former is 2% sample data and consists of the population sector as well as the household and housing sector. The population sector includes demographic variables such as age, sex, education level, marital status, and place of residence for individuals, as well as variables related to economic activity, such as occupation, industry, years of service at current job, place of work, mode of transportation, time taken to commute, etc. The household and housing sector includes variables related to the length of time spent in the currently occupied home and the type of housing tenure. The data can be integrated using household identification. The latter is annually published by KOTI and approved by Statistics Korea, in order to improve passenger and freight mobility and to secure the optimal transportation facilities that are necessary to support socioeconomic activities. It consists of the average access time, the accessible population ratio, and the number of accessible facilities to the service facilities, which is measured by car, public transportation, and walking, and indexed by time zone (daily average, morning peak, daytime, and evening peak). The average access time is the required time on average to reach the nearest service facility in each administrative district from the center of the administrative district. The accessible population ratio is the percentage of users who can be reached within a specific time period (15, 30, 45, 60 min) compared to the total population by region. The number of accessible facilities is the average number of service facilities that can be reached within a specific time period (15, 30, 45, 60 min) by region. Service facilities, a measured target in developing the transportation accessibility index, are largely divided into four types—educational facilities (elementary schools, middle schools, and high schools), medical facilities (public medical facilities, clinics, and general hospitals), sales facilities (large marts and traditional markets), and metropolitan transportation facilities (bus terminals, railway stations, and airports). In this study, we used transportation choices for educational facilities from the transportation accessibility index.

3.2. Models for Empirical Analyses

As mentioned above, the hypothesis of this paper is that differences in accessibility to educational facilities within a city affect the choice of commuting transportation for commuters in households with school-age children, and the binary logit model is used

to verify this. First, the utility of a particular mode (i.e., public transit) is assumed to be a function of the commuter's attributes and the accessibility index, as follows:

$$U_{ij} = \alpha + \gamma_1 age_{ij} + \gamma_2 edu_{ij} + \gamma_3 emp_{ij} + \gamma_4 MS_{ij} + \gamma_5 ind_{ij} + \gamma_6 job_{ij} + \gamma_7 DE_{ij} + \gamma_8 child_{ij} + \gamma_9 student_{ij} + \gamma_{10} commuter_{ij} + \gamma_{11} DR_{ij} + \gamma_{12} type_{ij} + \beta \Delta T_j$$

Then, the probability of using a public transit is given as

$$p = e^{U_{ij}} / \left(1 + e^{U_{ij}}\right) \quad (1)$$

The attributes of the commuters in question are controlled for as follows: age (age_{ij}); education (edu_{ij}); employment status (emp_{ij}); marital status (MS_{ij}); industry (ind_{ij}); occupation (job_{ij}); duration of employment at the current job (DE_{ij}); number of preschool children ($child_{ij}$); number of elementary, middle, and high school students in the household ($student_{ij}$); number of commuters ($commuter_{ij}$); duration of residence in the current house (DR_{ij}); and type ($type_{ij}$) of the housing unit. ΔT_j refers to the difference in average access time to educational facilities across transport modes, which was measured during peak morning travel hours (between 7:00 and 9:00). This is defined as the access time when using public transportation minus the access time when using a private car, and is measured by dividing it into elementary, middle, and high schools. The hypothesis of this paper implies that β has a negative (−) value. For the estimation of Equation (1), Stata 16 was used.

3.3. Variables

Table 1 reports summary statistics of variables. Approximately 32.3% of the sample uses public transportation as a commuting mode. On average, the sample population is aged 45 years, with age ranging between 17 and 85 years. Around 61.5% are college-educated or higher. In total, 63.8% of the sampled population have a spouse. One-fifth of the sampled population belong to the category of "professionals and related workers" and another one-fifth are classified as clerks. The share of the sampled population with school kids is 2.9% for elementary school, 8.7% for middle school, and 8.6% for high school. There are slightly less than two commuters in each household. A total of 59.5% of samples live in apartments, which is typical in Korea.

Table 1. (a) Summary statistics on the characteristics of individual commuters. (b). Summary statistics on the commuting time of areas (i.e., cities, counties, and wards).

Variables	Mean	Std. Dev.	Min.	Max.
(a)				
Use of public transit (1 for city bus, express/intercity bus, subway/train; 0 for passenger car or small van)	0.323	0.468	0	1
Age	45.007	12.546	17	85
Educational attainments				
No schooling	0.001	0.032	0	1
Elementary school	0.016	0.125	0	1
Middle school	0.044	0.206	0	1
High school	0.324	0.468	0	1
College	0.180	0.384	0	1
University	0.357	0.479	0	1
Master's	0.060	0.238	0	1
PhD	0.018	0.133	0	1
Employment status				
Paid worker	0.823	0.382	0	1
Self-employed w/o employees	0.106	0.308	0	1
Self-employed w/ employees	0.052	0.222	0	1
Non-paid family worker	0.019	0.138	0	1

Table 1. *Cont.*

Variables	Mean	Std. Dev.	Min.	Max.
(a)				
Marital status				
Single	0.277	0.447	0	1
Have a spouse	0.638	0.481	0	1
Widowed	0.023	0.149	0	1
Divorced	0.062	0.241	0	1
Occupation				
Managers	0.009	0.096	0	1
Professionals and related workers	0.218	0.413	0	1
Clerks	0.228	0.42	0	1
Service workers	0.118	0.322	0	1
Sales workers	0.109	0.312	0	1
Agriculture, forestry, and fisheries skilled workers	0.021	0.143	0	1
Craft and related trades workers	0.091	0.287	0	1
Plant and machine operators and assemblers	0.117	0.321	0	1
Elementary workers	0.082	0.275	0	1
Armed forces	0.007	0.086	0	1
Duration in current job				
Less than 6 months	0.071	0.256	0	1
6 months to 1 year	0.070	0.255	0	1
1 to 3 years	0.154	0.361	0	1
3 to 5 years	0.122	0.327	0	1
5 to 10 years	0.178	0.383	0	1
10 to 15 years	0.145	0.352	0	1
15 to 20 years	0.084	0.278	0	1
At least 20 years	0.176	0.381	0	1
Number of pre-school kids	0.158	0.456	0	5
Number of elementary school kids	0.029	0.171	0	2
Number of middle school kids	0.087	0.309	0	3
Number of high school kids	0.086	0.303	0	4
Number of commuters	1.782	0.789	1	11
Length of residing in current housing unit				
Less than 1 year	0.138	0.345	0	1
1 to 2 years	0.127	0.333	0	1
2 to 3 years	0.102	0.302	0	1
3 to 5 years	0.151	0.358	0	1
5 to 10 years	0.186	0.389	0	1
10 to 15 years	0.116	0.321	0	1
15 to 20 years	0.063	0.244	0	1
20 to 25 years	0.051	0.22	0	1
At least 25 years	0.065	0.247	0	1
Type of housing unit				
Single-detached	0.238	0.426	0	1
Apartment	0.595	0.491	0	1
Townhouse	0.025	0.157	0	1
Multi-generational housing	0.095	0.293	0	1
Unit in non-residential building	0.010	0.101	0	1
Officetel	0.031	0.173	0	1
Room in a hotel, inn, or other lodging facility	0.001	0.025	0	1
Dormitory and social facilities	0.000	0.017	0	1
Shack or vinyl house	0.000	0.015	0	1
Others	0.003	0.058	0	1
(b)				
Elementary school kids by private car	4.2	1.7	2.3	15.6
Elementary school kids by public transit	12.0	8.4	4.4	45.6
Middle school kids by private car	5.6	2.5	3.1	22.5
Middle school kids by public transit	16.6	10.5	6.1	57.4
High school kids by private car	8.0	8.7	3.9	120
High school kids by public transit	20.0	13.2	6.9	120

The commuting time of school kids seems to vary remarkably. The older the students are, the longer it takes for their commute to school. Also, the difference in commuting time between transportation modes is large, no matter which school the students attend. It takes more time when they commute to school using public transportation by 7.8 mins for elementary school, 11.0 mins for middle school, and 12.0 mins for high school.

4. Empirical Results and Discussion

4.1. The Probability of Using Public Transportation as a Mode of Commuting

Before examining the influence of urban traffic conditions on the choice of commuting transportation, we analyze whether the differences in the choice of commuting transportation by commuter attributes are different between men and women. That is, we estimate a model in which ΔT_j is not included in Equation (1); Table 2 shows the results. The first column shows the results for female commuters, and the last column shows the results for male commuters. For women, there is no significant difference in the probability of using public transportation by age; however, for men, the probability of using public transportation increases with age. As the level of education increases, women generally become less likely to use public transportation, but for men, the likelihood of using public transportation is lowest when the highest level of education is high school. In terms of differences by occupation, women are most likely to use public transport if they are craft and related trades workers, while men are most likely to use public transport if they are professionals and related workers, showing differences between the sexes.

Table 2. Estimation results on the probability of using public transit as a commuting mode.

Variables	Female	Male
Age	0.001	0.007 **
	(−1.22)	(−8.65)
Education (reference: no schooling)		
Elementary school	−0.492 **	−0.754 **
	(−3.12)	(−2.81)
Middle school	−1.017 **	−1.142 **
	(−6.56)	(−4.29)
High school	−1.462 **	−1.444 **
	(−9.48)	(−5.45)
College	−1.584 **	−1.519 **
	(−10.19)	(−5.72)
University	−1.531 **	−1.196 **
	(−9.86)	(−4.51)
Master's	−1.721 **	−1.094 **
	(−10.95)	(−4.11)
PhD	−1.697 **	−1.233 **
	(−10.28)	(−4.60)
Employment status (reference: paid worker)		
Self-employed without employees	−0.633 **	−0.558 **
	(−26.49)	(−24.63)
Self-employed with employees	−1.035 **	−1.011 **
	(−28.19)	(−30.98)
Non-paid family worker	−0.695 **	−0.116 +
	(−16.86)	(−1.91)
Marital status (reference: single)		
Have a spouse	−1.048 **	−0.764 **
	(−52.09)	(−40.87)
Widowed	−0.491 **	−0.490 **
	(−13.09)	(−7.40)
Divorced	−0.753 **	−0.540 **
	(−26.44)	(−17.32)

Table 2. *Cont.*

Variables	Female	Male
Occupation (reference: manager)		
Professionals and related workers	0.724 **	0.534 **
	(−6.77)	(−8.66)
Clerks	0.761 **	0.356 **
	(−7.09)	(−5.76)
Service workers	0.627 **	0.344 **
	(−5.8)	(−5.14)
Sales workers	0.748 **	0.291 **
	(−6.85)	(−4.48)
Agriculture, forestry, and fisheries skilled workers	0.106	−0.283 *
	(−0.57)	(−2.35)
Craft and related trades workers	1.027 **	0.186 **
	(−9.07)	(−2.90)
Plant and machine operators and assemblers	0.556 **	0.053
	(−4.95)	(−0.83)
Elementary workers	0.962 **	0.376 **
	(−8.81)	(−5.84)
Armed forces	−2.226 **	−1.483 **
	(−5.07)	(−11.11)
Duration on current job (reference: less than six months)		
6 months to 1 year	−0.277 **	−0.222 **
	(−9.56)	(−7.49)
1 to 3 years	−0.290 **	−0.399 **
	(−11.62)	(−15.67)
3 to 5 years	−0.393 **	−0.569 **
	(−14.90)	(−21.19)
5 to 10 years	−0.432 **	−0.586 **
	(−17.15)	(−22.90)
10 to 15 years	−0.476 **	−0.578 **
	(−17.63)	(−21.51)
15 to 20 years	−0.586 **	−0.617 **
	(−18.21)	(−20.46)
At least 20 years	−0.655 **	−0.662 **
	(−21.65)	(−24.27)
Number of kids in the household		
Pre-school	−0.239 **	−0.069 **
	(−13.76)	(−4.58)
Elementary school	−0.258 **	−0.082 *
	(−6.79)	(−2.17)
Middle school	−0.228 **	−0.141 **
	(−10.76)	(−6.36)
High school	−0.120 **	−0.068 **
	(−5.85)	(−3.13)
Number of commuters in the household	0.133 **	0.143 **
	(−16.04)	(−18.71)
Duration of residing in the current household (ref.: less than 1 year)		
1 to 2 years	−0.062 **	0.03
	(−2.60)	(−1.34)
2 to 3 years	−0.101 **	0.008
	(−4.00)	(−0.34)
3 to 5 years	−0.085 **	0.093 **
	(−3.70)	(−4.32)
5 to 10 years	−0.143 **	0.006
	(−6.46)	(−0.27)
10 to 15 years	−0.161 **	−0.012
	(−6.51)	(−0.49)
15 to 20 years	−0.074 *	0.089 **
	(−2.52)	(−3.12)

Table 2. *Cont.*

Variables	Female	Male
20 to 25 years	−0.012	0.018
	(−0.38)	(−0.57)
At least 25 years	−0.351 **	−0.197 **
	(−11.18)	(−6.44)
Type of housing unit (reference: single-detached)		
Apartment	−0.047 **	−0.013
	(−3.03)	(−0.87)
Townhouse	0.135 **	0.120 **
	(−3.49)	(−3.11)
Multi-generational housing	0.849 **	0.642 **
	(−36.15)	(−29.89)
Unit in non-residential building	0.401 **	0.516 **
	(−6.86)	(−9.67)
Officetel	0.653 **	0.480 **
	(−17.55)	(−14.42)
Room in a hotel, inn, or other lodging facility	1.438 **	1.012 **
	(−4.58)	(−5.3)
Dormitory and social facilities	0.072	−0.182
	(−0.16)	(−0.52)
Shack or vinyl house	−0.639 +	−0.871
	(−1.72)	(−1.32)
Others	−0.382 **	0.390 **
	(−3.33)	(−4.15)
Constant	0.169	−0.723 *
	(−0.72)	(−2.45)
Observations	132,112	185,691
Pseudo R-squared	0.127	0.108
Log-L	−79,303	−90,385

Note. Robust t-values are in parentheses. Error terms are clustered at the level of city, county, and ward. **, *, and + indicate statistical significance at 1%, 5%, and 10%, respectively. Dummy variables for individual industries are included in the regression, but due to a limit on space, their results are suppressed.

Meanwhile, in terms of employment status, both men and women are most likely to use public transportation if they are salaried workers. In terms of marital status, the proportion of unmarried people commuting to work by public transportation is the highest, and this is true for both men and women. Married females are less likely to use public transportation than single females by 21.7 percentage points while married males are less likely to use public transportation than single males just by 12.1 percentage points. The longer the tenure at the current job, the lower the probability of using public transportation, and this was observed regardless of gender. The presence of children in the household appears to lower the likelihood of commuters using public transportation, which is a finding that seems reasonable given that it may be more convenient to commute by car when dropping off or picking up children from kindergarten or school on the way to or from work. This is seen for both female and male commuters; however, in the case of women, the probability of using public transportation is lower, which is interpreted as meaning that the burden of children's school-related transportation is mainly borne by women (i.e., wives). There is no monotonic relationship observed between the duration of residence in the current residence and the probability of using public transportation; for male commuters, the significance by period is generally low.

4.2. Differences in Average Commuting Times by City

Table 3 shows the results for ΔT_j from Equation (1), which includes the average commuting time by public transportation in the region j minus the average commuting time by private car. Column (1) depicts the results of defining ΔT_j, which is the time

required for elementary school students to commute to school. The longer it takes for elementary school students in a given area to commute to school by public transportation compared to driving, the more likely it is that parents will drive their children to or from school. Columns (2) and (3) correspond to the results defining ΔT_j, which is the commuting time for middle school and high school students, respectively. This is similar to the results in column (1) targeting elementary school students. However, the younger the child is, the greater the difference in average commute times between modes of transportation in the area of residence, which appears to have an impact on the likelihood of using public transportation. Figure 1 visualizes the key findings in Table 3.

Table 3. Choice of public transit as a commuting mode—difference in travel time of school kids by mode.

Variables	(1)	(2)	(3)
ΔT_j			
Elementary school	−0.386 **		
	(−6.40)		
Middle school		−0.266 **	
		(−9.61)	
High school			−0.241 **
			(−11.59)
Observations	9179	24,987	25,156
Pseudo R-squared	0.15	0.156	0.175
Log-L	−4126	−11,264	−11,756

Note. Robust t-values are in parentheses. Error terms are clustered at the level of city, county, and ward. ** indicate statistical significance at 1%. The attributes of individual commuters were included in the regression, but their coefficients are suppressed due to limited space.

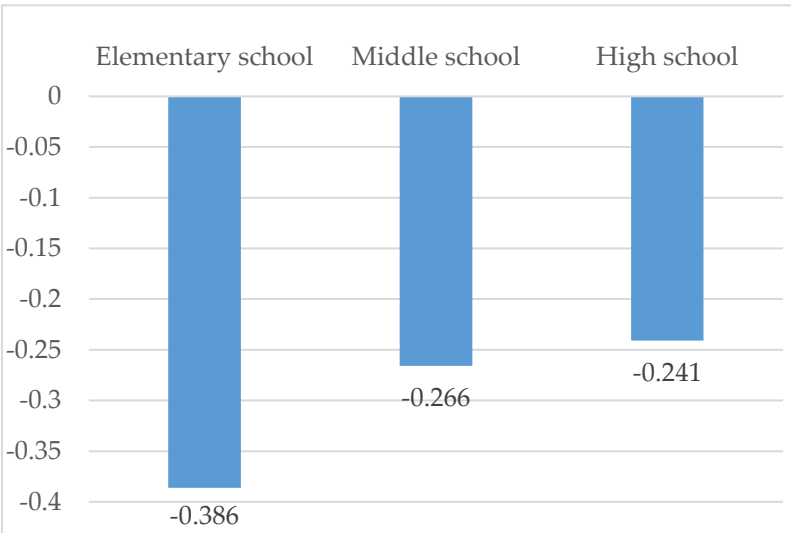

Figure 1. Impact of difference in travel time of school children by mode on the choice of public transit.

Similarly to Table 3, Table 4 shows the results of ΔT_j from Equation (1). However, the estimation was performed separately for wives and husbands. In the first two columns which are for households with elementary school students, both wives and husbands tend to use private transportation as a commuting mode rather than public transit in a city where students' traveling to schools by public transit takes more time than by private transit. But this tendency is stronger with husbands than wives. In the third and fourth columns for households with middle school students, similar results appear. However, the difference between wives and husbands is greater than the households with elementary

school students. The last two columns for households with high school students show the largest difference between wives and husbands. The results in Table 4 confirm those in Table 3; that is, parents would care more for their children's commuting as they are younger. Wives seem to take more responsibility of young children's commuting than husbands in cities where the infrastructure of public transit is less developed [33]. As the children advance to a higher grade, wives tend to be free from the responsibility and are less reluctant to use public transit. Figure 2 summarizes the key results in Table 4.

Table 4. Choice of public transit as a commuting mode—difference in childbearing burden among couples.

Variables	(1)		(2)		(3)	
	Wife	Husband	Wife	Husband	Wife	Husband
ΔT_j						
Elementary school	−0.392 **	−0.404 **				
	(−6.09)	(−4.24)				
Middle school			−0.313 **	−0.338 **		
			(−7.29)	(−5.35)		
High school					−0.289 **	−0.327 **
					(−8.33)	(−7.13)
Observations	3816	5360	3816	5360	3816	5360
Pseudo R-squared	0.171	0.158	0.186	0.172	0.192	0.183
Log-L	−1910	−2069	−1875	−2034	−1862	−2009

Note. Robust t-values are in parentheses. Error terms are clustered at the level of city, county, and ward. ** indicate statistical significance at 1%. The attributes of individual commuters were included in the regression, but their coefficients are suppressed due to limited space.

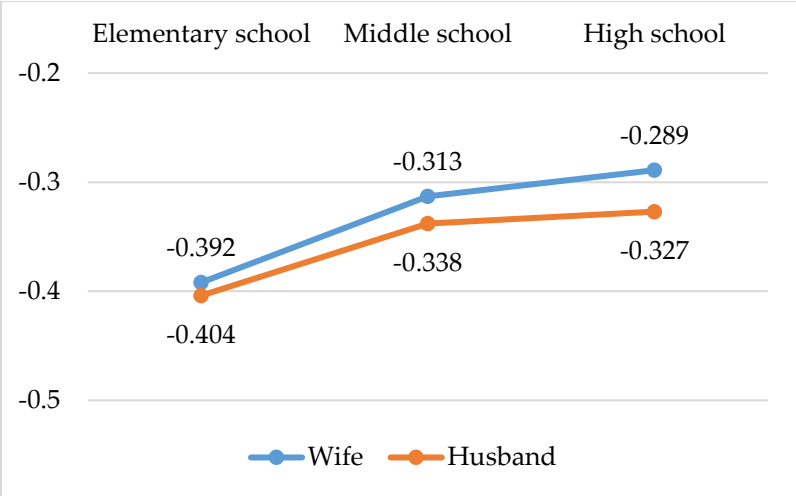

Figure 2. Impact of difference in childbearing burden among couples on the choice of public transit.

5. Discussion

When there is a large difference between transportation modes in terms of accessibility to educational facilities within the residential area, it is observed that this affects parents' choice of transportation mode for commuting. In particular, as accessibility to public transportation decreases, the proportion of commuting by private car increases, which reduces the demand for public transportation and fare revenue in the area, increasing the financial burden on local governments to provide local public transportation services. In addition, the increased use of passenger cars has a negative impact on the air quality in the area. In Korea, parents' enthusiasm for their children's education has increased significantly compared to in the past due to the continued low birth rate; similarly, women's participation in the labor market has increased due to improved educational attainment and changes in

social awareness. These changes indicate that the presence of school-age children in the household significantly affects married female commuters' use of public transportation to commute to work. The empirical analysis results of this paper suggest that improving public transportation and accessibility to educational facilities in a given area is important in increasing public transportation usage.

6. Conclusions

This paper empirically analyzed the hypothesis that differences in accessibility to educational facilities by commuting mode will affect the choice of commuting mode for parents with school-age children through a binary logit model using the Population and Housing Census of Statistics Korea and the transportation accessibility index of the Korea Transport Institute. Empirical analysis results show that the worse the accessibility to educational facilities by public transportation compared to private vehicles, the lower the likelihood that parents in the city will commute by public transportation. In addition, because the burden of child-rearing differs between husbands and wives and the distances to school differ between elementary-, middle-, and high-school-age children, different results emerged between couples and between children of a different school age. The younger the child, the more likely they are to commute to work using public transportation in areas with poorer relative accessibility to public transportation. Additionally, this decrease in the likelihood of using public transportation appears to be greater for fathers of school-aged children than for mothers. This is believed to be due to the fact that married women have relatively lower wages than their husbands due to career gaps (for example, secondary earners in dual-income couples) or that they commute relatively short distances due to the burden of raising children; therefore, they use public transportation more often and it is not easy to switch to using private cars. From a policy perspective, improving public transportation access to educational facilities in cities will contribute to the soundness of public finances for maintaining public transportation services and to improving the sustainability of cities through improved air quality and alleviating traffic congestion. Therefore, expanding public transportation infrastructure in cities remains important.

In Korea, the use of private transportation by female workers has been increasing. Thus, it would be meaningful to compare results over different time periods. As the role of female workers in the Korean labor market has grown, they are often promoted to high positions and earn high wages, which allows them to not be dependent on public transit. However, this is another topic that can be tackled in future work. Also, the results could have been affected by the COVID-19 pandemic, since the year of the data used in this paper is 2020. This issue could be addressed by comparing results over different time points.

Funding: This research received external funding from Chungnam Institute.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available from MDIS and KOSIS by Statistics Korea.

Acknowledgments: The author would like to thank the three anonymous reviewers for their valuable comments and suggestions on previous versions of this paper.

Conflicts of Interest: The author declares no conflicts of interest.

References

1. Lee, B.S.; McDonald, J.F. Determinants of Commuting Time and Distance for Seoul Residents: The Impact of Family Status on the Commuting of Women. *Urban Stud.* **2003**, *40*, 1283–1302. [CrossRef]
2. Lu, H.; Gan, H. Unraveling the Influence of Perceived Built Environment on Commute Mode Choice Based on Hybrid Choice Model. *Appl. Sci.* **2024**, *14*, 7921. [CrossRef]
3. McQuaid, R.W.; Chen, T. Commuting Times—The Role of Gender, Children and Part-Time Work. *Res. Transp. Econ.* **2012**, *34*, 66–73. [CrossRef]
4. Gimenez-Nadal, J.I.; Molina, J.A. Commuting Time and Household Responsibilities: Evidence Using Propensity Score Matching. *J. Reg. Sci.* **2016**, *56*, 332–359. [CrossRef]
5. Ortúzar, J.D.; Willumsen, L.G. *Modelling Transport*, 4th ed.; Wiley Press: Oxford, UK, 2011; ISBN 978-0-470-76039-0.
6. Truong, L.; Somenahalli, S.V.C. Exploring Frequency of Public Transport Use among Older Adults: A Study in Adelaide, Australia. *Travel Behav. Soc.* **2015**, *2*, 148–155. [CrossRef]
7. Pettersson, P.; Schmöcker, J.D. Active Ageing in Developing Countries?—Trip Generation and Tour Complexity of Older People in Metro Manila. *J. Transp. Geogr.* **2010**, *18*, 613–623. [CrossRef]
8. Jin, H.; Yu, J. Gender Responsiveness in Public Transit: Evidence from the 2017 US National Household Travel Survey. *J. Urban Plan. Dev.* **2021**, *147*, 04021021. [CrossRef]
9. Nguyen, T.M.C.; Kato, H.; Phan, L.B. Is Built Environment Associated with Travel Mode Choice in Developing Cities? Evidence from Hanoi. *Sustainability* **2020**, *12*, 5773. [CrossRef]
10. Batabyal, A.A.; Beladi, H. Commuting to Work in Cities: Bus, Car, or Train? *Reg. Sci. Policy Pract.* **2022**, *13*, 599–610. [CrossRef]
11. Hjorthol, R.; Vågane, L. Allocation of Tasks, Arrangement of Working Hours and Commuting in Different Norwegian Households. *J. Transp. Geogr.* **2014**, *35*, 75–83. [CrossRef]
12. Wheatley, D. Travel-to-Work and Subjective Well-Being: A Study of UK Dual Career Households. *J. Transp. Geogr.* **2014**, *39*, 187–196. [CrossRef]
13. Campaña, J.C.; Gimenez-Nadal, J.I. Gender Gaps in Commuting Time: Evidence from Peru, Ecuador, Chile, and Colombia. *J. Fam. Econ. Iss.* **2024**, *45*, 596–620. [CrossRef]
14. Marcén, M.; Morales, M. Culture and the Cross-Country Differences in the Gender Commuting Gap. *J. Transp. Geogr.* **2021**, *96*, 103184. [CrossRef]
15. Lejsková, P.; Pojkarová, K.; Kudláčková, N.; Becková, H.; Čubranić-Dobrodolac, M. Gender Differences in Transport Behaviour Patterns. *LOGI–Sci. J. Transp. Logist.* **2023**, *14*, 329–340. [CrossRef]
16. Liu, Y.; Min, S.; Shi, Z.; He, M. Exploring Students' Choice of Active Travel to School in Different Spatial Environments: A Case Study in a Mountain City. *J. Transp. Geogr.* **2024**, *115*, 103795. [CrossRef]
17. Wang, X.; Shao, C.; Yin, C.; Guan, L. Built Environment, Life Events and Commuting Mode Shift: Focus on Gender Differences. *Transp. Res. Part D Transp. Environ.* **2020**, *88*, 102598. [CrossRef]
18. Hayauchi, G.; Nakamura, F.; Ariyoshi, R.; Tanaka, S.; Miura, S. A Study on Influence of Topographical Factors to Mode Choice. *J. Jap. Soc. Civil Eng.* **2019**, *75*, 565–574. [CrossRef]
19. Sun, B.D.; Ermagun, A.; Dan, B. Built Environmental Impacts on Commuting Mode Choice and Distance: Evidence from Shanghai. *Transp. Res. Part D Transp. Environ.* **2017**, *52*, 441–453. [CrossRef]
20. Hasan, A. Development of Intercity Mode Choice Models for Saudi Arabia. J. King Abdulaziz univ. *Eng. Sci.* **2006**, *17*, 3–21. [CrossRef]
21. Assi, K.J.; Nahiduzzaman, K.M.; Ratrout, N.T.; Aldosary, A.S. Mode Choice Behavior of High School Goers: Evaluating Logistic Regression and MLP Neural Networks. *Case Stud. Transp. Policy* **2018**, *6*, 225–230. [CrossRef]
22. Stewart, O.; Moudon, A.V.; Claybrooke, C. Common Ground: Eight Factors that Influence Walking and Biking to School. *Transp. Policy* **2012**, *24*, 240–248. [CrossRef]
23. Zhang, R.; Yao, E.; Liu, Z. School Travel Mode Choice in Beijing, China. *J. Transp. Geogr.* **2017**, *62*, 98–110. [CrossRef]
24. Singh, N.; Vasudevan, V. Understanding School Trip Mode Choice—The Case of Kanpur (India). *J. Transp. Geogr.* **2018**, *66*, 283–290. [CrossRef]
25. Ahern, S.M.; Arnott, B.; Chatterton, T.; De Nazelle, A.; Kellar, I.; McEachan, R.R.C. Understanding Parents' School Travel Choices: A Qualitative Study Using the Theoretical Domains Framework. *J. Transp. Health* **2017**, *4*, 278–293. [CrossRef]
26. Fitzhugh, E.C.; Everett, J.; Daugherty, L. What Parental Correlates Predict Children's Active Transportation to School in the Southeast United States? *J. Phys. Act. Health* **2021**, *18*, 705–713. [CrossRef]
27. Guthrie, A.; Fan, Y. Weakening Obstacles to Transit Use: Changes in Relationships with Child Rearing and Automobile Access from 2000 to 2010. *Transp. Res. Rec.* **2016**, *2561*, 103–110. [CrossRef]
28. Echeverria, L.; Gimenez-Nadal, J.; Molina, J.A. Commuting in Dual-Earner Households: International Gender Differences with Time Use Surveys. *Rev. Econ. Househ.* **2024**, 1–23. [CrossRef]

29. Yang, H.C.; Jin, L.; Lazar, A.; Todd-Blick, A.; Sim, A.; Wu, K.; Chen, Q.; Spurlock, C. Gender Gaps in Mode Usage, Vehicle Ownership, and Spatial Mobility When Entering Parenthood: A Life Course Perspective. *Systems* **2023**, *11*, 314. [CrossRef]
30. Borghorst, M.; Mulalic, I.; Van Ommeren, J. Commuting, Children and the Gender Wage Gap. In *Tinbergen Institute Discussion Paper*; 2021; TI 2021-089/VIII, pp. 1–48. Available online: https://papers.tinbergen.nl/21089.pdf (accessed on 22 April 2025).
31. Borghorst, M.; Mulalic, I.; Ommeren, J. Commuting, Gender and Children. *J. Urban Econ.* **2024**, *144*, 103709. [CrossRef]
32. Ma, S.; Yin, X.; Tang, D.; Liu, C. An Analysis of Residents' Commuting Behavior Considering Household Heterogeneity. *J. Phys. Conf. Ser.* **2019**, *1168*, 032100. [CrossRef]
33. Liu, Y.; Shen, R.; He, M.; Li, X.; Shi, Z. Gender Differences in Commuting Travel Mode Choices among Young Adults: A Spatial Heterogeneity Perspective. *J. Transp. Geogr.* **2025**, *123*, 104145. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.